

Homework**Polona Oblak**

Let m be the last digit in your Student ID that is not from the set $\{0, 1, 2\}$, and define $f_1, f_2, f_3: \mathbb{R}_2[x] \rightarrow \mathbb{R}$ as

$$f_1(p) = p(-m) + p(m),$$

$$f_2(p) = p'(m) + p''(0),$$

$$f_3(p) = \int_{-1}^1 p(x) dx,$$

and let $\varphi: \mathbb{R}_2[x] \times \mathbb{R}_2[x] \rightarrow \mathbb{R}$ as

$$\varphi(p, q) = \int_0^1 p(x)q(x) dx.$$

- (1) Take a moment to think why $f_1, f_2 \in (\mathbb{R}_2[x])^*$. No need to prove it, this part will not be graded.
- (2) Prove that $f_3 \in (\mathbb{R}_2[x])^*$.
- (3) Prove that $\{f_1, f_2, f_3\}$ form a basis for $(\mathbb{R}_2[x])^*$.
- (4) Find the basis $\mathcal{B} = \{p_1, p_2, p_3\}$ for $\mathbb{R}_2[x]$ such that $\mathcal{B}^* = \{f_1, f_2, f_3\}$.
- (5) Show that φ is bilinear form on $\mathbb{R}_2[x]$ and express φ as a linear combination of $\{f_i \otimes f_j: i, j = 1, 2, 3\}$. Write down all nine coefficients.

The solution of the problem needs to be submitted to the appropriate mailbox on Ucilnica until a day before the exam, but preferably in a week. The solutions should be written by hand (on paper and scanned or on a tablet) and should contain clear and well-explained proofs, procedures, explanations, etc. At the end, add a short statement confirming that the submitted solution is your own work. Also list the names of people you discussed the assignment with (if any) and clearly indicate which parts of the solution were developed by you and which parts were assisted by AI tools (if any).